Circuit-level equivariance is code-dependent: obstruction in the surface code, exactness in the honeycomb Floquet code, and consequences for neural decoders

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Abstract

Symmetry-equivariant neural decoders for quantum error correction were introduced by Egorov, Bondesan and Welling for the toric code at code capacity, where the optimal decoder's symmetry group — including syndrome-dependent corrections to the logical labels — can be derived analytically from lattice paths. Real decoders, however, operate at circuit level, on the detector error model (DEM) generated by a syndrome-extraction circuit. We show that whether exact equivariance survives the passage to circuit level is a property of the *code*. For the surface code it does not: an exhaustive search over 999 affine spacetime detector maps and a constructive co-design search over 1,728 extraction schedules find no exact twisted symmetry of the circuit-level DEM within the geometric family, and we identify the mechanism — a mirror-axis ancilla would need a CNOT order equal to its own reflection, which no order can be. For the honeycomb Floquet code an exact $p3$ space group — the color-preserving translations and C_3 rotations — survives even naive ancilla-compiled extraction (the color-cycling elements survive as exact spacetime screws in the time bulk): trivalence makes every check weight-2, reducing schedule freedom to one bit per edge (automatically covariant) and hook errors to weight one. We solve the symmetry action on the logical observables directly from the DEM as a triple (P, A, X) — detector permutation, $GL(2, \mathbb{F}_2)$ action on the two logical qubits, and a syndrome-dependent twist — verify it against every DEM mechanism group, and validate the solver by blind recovery of the known analytic action on the toric code. The C_3 rotation mixes the logical qubits (A of order 3), so the correct equivariant decoder is a four-class, jointly-equivariant model. On the honeycomb circuit-level task, a matched-parameter unconstrained network remains at the majority-class baseline through 64,000 training samples, while the exactly-tied model reaches 0.806 ± 0.003 at 256,000 samples — the tied model at 1,000 samples outperforms the unconstrained model at 256,000, a $\geq 256\times$ sample-efficiency separation. A noise-rate sweep finds the tied advantage at every rate tested and locates the strict switch in a noise window around the gated rate. On the surface code under phenomenological noise, where exact symmetry exists below circuit level, the group elements that *move* the logical representative (twisted elements) carry essentially all of the benefit. Code choice, not architecture choice, determines whether exact equivariance is available to machine-learned decoders where devices actually operate.

1 Introduction

Machine-learned decoders are now competitive with, and in some regimes ahead of, hand-crafted decoders for quantum error correction: AlphaQubit [1] set the accuracy state of the art on Sycamore surface-code data, and learned decoders adapt to real noise that analytic decoders model imperfectly. The binding constraint in that program is *data*: syndrome samples

from real hardware are scarce and expensive, which makes inductive bias — how much of the decoding function is built in rather than learned — the central design question.

The natural inductive bias for a topological code is its lattice symmetry. Egorov, Bondesan and Welling ("The END") [5] formalized this for the toric code at code capacity: the optimal decoder is equivariant under the code's automorphism group *provided* the output representation is twisted — the logical-class label transforms by a group-element-dependent permutation composed with a syndrome-dependent correction, because a symmetry maps a logical representative to a homologous representative that differs by stabilizers. Their equivariant architecture, built on a "twisted global average pooling," beat both MWPM and prior neural decoders, and — a result we will return to — their ablation found that *untwisted* pooling fails to learn at all.

The END's analysis lives at code capacity: perfect syndrome measurements, no extraction circuit, with the twist derived analytically from lattice paths. Real decoders consume the output of a *syndrome-extraction circuit*, and the object they actually decode is the circuit's detector error model (DEM): the hypergraph of elementary error mechanisms, the detectors they flip, and the logical observables they cross. Whether the lattice symmetry survives in the DEM is not guaranteed by the symmetry of the code: the extraction schedule — the order in which entangling gates touch each ancilla — imprints itself on the noise through hook errors, and schedules are discrete objects with no obligation to respect the lattice's point group.

This paper asks: **for which codes does exact (twisted) equivariance survive at circuit level — and what does the answer do to learned decoders?**

Our contributions:

1. **A DEM-native solver for twisted symmetry actions** (§3). Given any stim circuit and a candidate detector permutation, we solve for the action on the logical observables — a pair (A, X) with $A \in GL(k, \mathbb{F}_2)$ acting on the k logical bits and X a syndrome-linear twist — from the DEM alone, by per-detector-set multiset equations, and verify the result against every mechanism group exactly. No analytic path construction is needed, which is what makes the method applicable at circuit level. Blind-searched over all 24 label permutations on The END's own setting (toric code, code capacity), the solver uniquely recovers their analytic action — including the 90° rotation's logical swap and the weight-L stabilizer-path twists (§3.2).

2. **An obstruction result for the surface code** (§4). At circuit level, no exact twisted symmetry of the unrotated surface-code DEM exists in the geometric family: 999 affine spacetime maps (reflections, rotation, time-reversal, shifts) all fail verification, and a constructive search over 1,728 extraction-schedule assignments — including mirror-paired and time-reversed families designed specifically to restore symmetry — yields zero verified schedules. We identify the obstruction: an ancilla on the mirror axis would need a CNOT order whose hook-error prefixes equal their own mirror image, and no permutation equals its own nontrivial conjugate-reversal. Schedule co-design, recently an active topic for distance preservation [12, 6], cannot rescue exact equivariance for this code.

3. **An exactness result for the honeycomb Floquet code** (§5). The Hastings–Haah honeycomb code [10] compiled with *naive, uniform* ancilla extraction carries an exact $p3$ space group — color-preserving translations and C_3 face rotations — as a symmetry of its circuit-level DEM, verified by exact multiset equality over all 4,020 mechanism groups at $L = 6$; the color-cycling elements survive as exact time-bulk spacetime screws (composed with sub-round shifts). The mechanism is trivalence: weight-2 checks leave one bit of schedule freedom per edge, which is automatically covariant, and bound hook errors to weight one. No schedule engineering is involved.

4. **The logical action and its decoder consequence** (§5.3). Solving (A, X) on the honeycomb torus: translations act with $A = I$ (homology preserved, twist only), while C_3 acts with $A = \left[\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ — an order-3 element of $GL(2, \mathbb{F}_2)$ that *mixes the two logical qubits*. A scalar-label equivariant decoder therefore cannot express the code's rotation symmetry at all; the correct formulation is a four-class (joint two-bit) twisted-equivariant model, $f(x)[c] = \frac{1}{|G|} \sum_g \mathrm{net}(\tilde{x}_g)[A_g c \oplus X_g \cdot \tilde{x}_g]$. En route we find that the deterministic logical wraps of an X-basis-initialized honeycomb memory exist only at one termination phase in three (final readout in the Y basis, $S \equiv 1 \bmod 3$) — an operational fingerprint of the code's logical-type rotation [10] that our observable-derivation procedure discovers automatically.

5. **Learning experiments** (§6). (i) Surface code, phenomenological noise: at matched parameters the twisted-invariant decoder dominates plain at every training-set size (d=3: +4.2 pp at n=250, $\approx 4\times$ sample efficiency; d=5: Klein-four reaches at n=4,000 what plain reaches at 64,000, $>16\times$), and an element ablation shows the *twisted* elements — those that move the logical representative — carry essentially all the benefit, while the $\chi = 0$ mirror contributes almost nothing. (ii) Honeycomb code, circuit-level noise: the unconstrained matched-parameter MLP never leaves the majority-class baseline through 64,000 samples; the $p3$-tied model reaches 0.744 at 64,000 and 0.806 at 256,000, and the tied model at 1,000 samples exceeds the unconstrained model at 256,000 ($\geq 256\times$). Exact equivariance here is not an efficiency multiplier but a learnability switch — consistent with, and sharpening, The END's untwisted-pooling ablation; a noise-rate sweep (§6.2) bounds the strict switch to a window around the gated rate while the tied advantage holds at every rate. (iii) Composing a translation-equivariant CNN backbone with the twisted averaging improves the scarce-data regime further (+3 to +6 pp over the tied MLP at 10^3–10^4 samples) and isolates a structural fact: convolutional weight sharing cannot absorb the label twist — on the same conv backbone, explicit averaging over the very translations the convolution already "has" adds +21 pp, because the twist is a cocycle on the output, not a feature symmetry (§6.4).

We close with what this does and does not imply for practical decoding (§7). Throughout, "exact" means exact multiset equality of DEM mechanism groups under the candidate action — a verification standard, not an approximation.

2 Background

2.1 Detector error models and decoding as classification

A stabilizer memory experiment compiles to a Clifford circuit with noise; a *detector* is a parity of measurement records that is deterministic in the noiseless circuit, and a *logical observable* is such a parity that reads out encoded information. The detector error model extracted from the circuit (we use stim [7]) is a list of elementary mechanisms (D_i, o_i, p_i): a set of flipped detectors, a mask of flipped observables, and a probability. Decoding is the classification problem syndrome $\to$ observable mask; MWPM [4, 11] is the standard baseline; learned decoders [1] fit the map from samples.

2.2 Twisted equivariance (after The END)

A spatial symmetry g of the code permutes detectors by some P_g. It does not in general fix the chosen logical representatives; it maps them to homologous representatives differing by stabilizer products, whose measured values telescope into detector sums. The label therefore

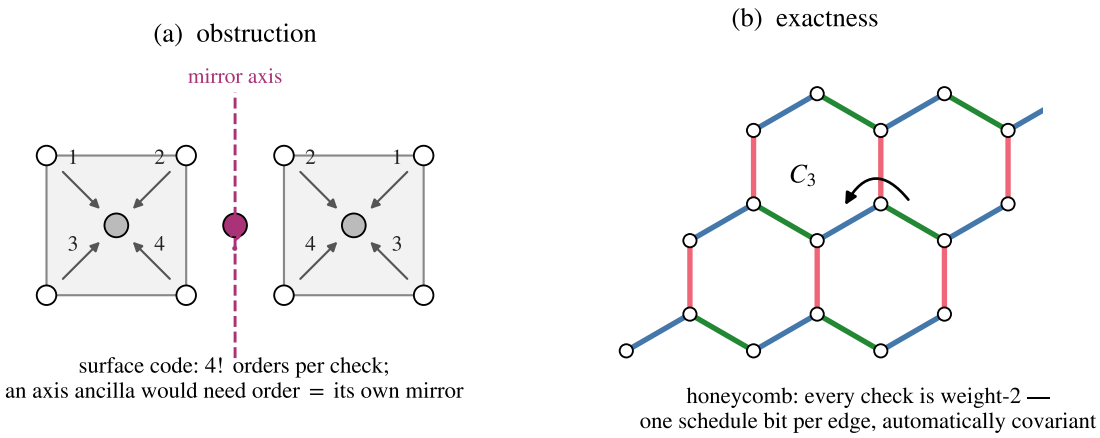


Figure 1: **Why circuit-level equivariance is code-dependent.** (a) Surface code: each weight-4 check admits 4! CNOT orders; hook-error prefixes depend on the order, and an ancilla on a mirror axis would need an order equal to its own reflection, which no order is (§4). (b) Honeycomb Floquet code: trivalence makes every check weight-2, leaving one schedule bit per edge; a uniform sublattice rule is automatically covariant, so an exact $p3$ space group survives in the circuit-level DEM under naive extraction (§5).

transforms as $o \to A_g\, o \oplus X_g \cdot (P_g s)$ with A_g a permutation-like linear action on the logical bits and X_g a fixed $\mathbb{F}_2$ matrix applied to the permuted syndrome s. At code capacity on the toric code, (A, X) can be written down from lattice paths [5]. The decoder respecting this symmetry is the group-averaged, twist-corrected model used throughout this paper.

2.3 Codes and circuits

Surface code: stim's unrotated planar memory (Z basis), distances 3–11, phenomenological noise for §6.1 (measurement-flip + data depolarization, $p = 0.008$) and full circuit-level noise for §4.

Honeycomb Floquet code [10]: torus, $L \times M$ cells (72 data qubits at $L = 6$), faces 3-colored, sub-round s measures color-(s mod 3) edges in Pauli basis $(X, Y, Z)[s \bmod 3]$; detectors are face inference windows compared across periods. We compile three extraction modes — direct two-body measurement (`mpp`), naive ancilla extraction with a uniform "A-sublattice-first" gate order (`cx-naive`), and an orbit-equivariant order (`cx-sym`) — and find (§5) that the third is unnecessary: naive extraction is already exactly symmetric. Circuit-level noise: single-qubit depolarization on data each sub-round, two-qubit depolarization after each entangling gate, ancilla preparation/readout flips, $p = 0.002$ for the decoding experiments (chosen so the MWPM baseline operates in a learnable band; the noise-rate gate is described in §6.2).

3 Solving twisted symmetry actions from the DEM

3.1 The per-group multiset method

Group the DEM by detector set: by_v$[D]$ = multiset of (o, p) over mechanisms with detector set D. A candidate action (P, A, X) is an exact symmetry iff for every D, the multiset $\{(Ao \oplus X{\cdot}\mathbf{1}_{P(D)},\, p)\}$ equals by_v$[P(D)]$. Because X enters only through the bit $b(D) = X{\cdot}\mathbf{1}_{P(D)}$, each detector-set group yields a small consistency problem: which label-offsets b make the multisets

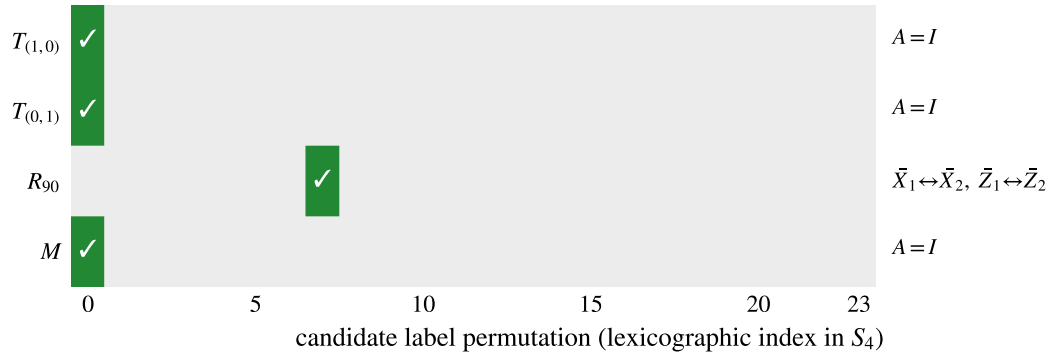


Figure 2: **Blind validation on The END's setting** (toric code, code capacity, $L = 6$). For each lattice generator the (A, X) solver is run against all 24 candidate label permutations; exactly one verifies (green), and it matches the analytic action of [5], including the 90°-rotation logical swap. Data: `runs/toric_end_bridge.json`.

match. Groups with a unique consistent b contribute linear equations $X \cdot \mathbf{1}_{P(D)} = b$; groups consistent with several b's are unconstrained; any group with none rejects (P, A). Gaussian elimination over $\mathbb{F}_2$ solves the resulting system row-wise; a final pass re-verifies every group against the solved action. Two practical notes: (i) stim merges mechanisms with identical detector sets, which is why the equations must be extracted per *group multiset* rather than per mechanism — naive mechanism-pairing generates false equations from probability collisions at $d \geq 5$; (ii) X is unique only modulo directions invisible to every mechanism set, so equivariance is exact on realizable syndromes (all train/test data), which is the relevant statement for decoding.

3.2 Validation: blind recovery of the analytic toric action

On The END's setting — toric code, code capacity, i.i.d. depolarizing — we ran the solver with *no analytic input*, searching all 24 permutations of the four logical labels for each generator. For every element the verifying A is unique and matches the analytic action [5]: translations and the mirror fix the labels ($A = I$) with twist χ supported on a weight-L stabilizer path along the moved representative; the 90° rotation swaps $\bar{X}_1 \leftrightarrow \bar{X}_2$ and $\bar{Z}_1 \leftrightarrow \bar{Z}_2$ with a single weight-L path twist. (Run time: $\sim$1 s; Appendix D.) This validates the solver in the one setting with a known analytic answer, and licenses its use at circuit level, where none exists.

4 Surface code: circuit-level obstruction

Affine family (999 maps). At circuit level (full extraction circuits, stim-generated, $d = 3$ and 5) we tested every affine spacetime candidate $P(x, y, t) = (\sigma(x, y), \pm t + \Delta t)$ with $\sigma \in \{\text{mirror}_x, \text{mirror}_y, C_2, \text{identity}\}$, time reversal, and shifts $\Delta \in [-2, 2]^3$: **zero** candidates verify, at either distance. This family is exhaustive for the geometric question: any DEM symmetry implementing a spacetime isometry of the memory experiment must act on detector coordinates as exactly such an affine map (point-group element composed with shifts and possibly time reversal), so what lies outside the family is by construction the non-geometric automorphism sector, which carries no spatial weight-sharing interpretation and is scoped out below. The extraction schedule's hook errors de-symmetrize the DEM: a hook error is a weight-2 data error created mid-extraction whose support depends on the CNOT order, and the orders do not transform into each other under the lattice maps.

Schedule co-design (1,728 schedules). Could the schedule be *chosen* to restore symmetry? We searched three families at $d = 3$ (rounds = 3, full circuit noise) with hard gates: G1 no gate collisions, G2 deterministic detectors, G3 circuit distance preserved, G4 exact twisted-symmetry verification. Family A (uniform per-orientation orders, 576): 96 collision-free, 16 pass G1–G3, **0 pass G4**. Family B (mirror-paired halves, 576): all 576 collide — the center-column ancilla's step-k east target is exactly the mirrored neighbor's step-k west target. Family C (mirror-paired with time-reversed right half, 576): the collision is fixed by construction, 32 valid, 8 pass G1–G3, **0 pass G4**.

The obstruction. An ancilla lying *on* the mirror axis must have a CNOT order whose hook-error prefix sets are invariant under the mirror's east–west exchange; concretely the order would have to equal its own nontrivial reversal-conjugate, which no permutation does. This is why both engineered families fail at the axis. The recent schedule-design literature optimizes a different objective — keeping hook errors off the logical operators to preserve circuit distance [12, 6] — and succeeds at it; exact DEM equivariance is a strictly stronger demand, and for this code it is unachievable within the geometric family. (Non-geometric DEM automorphisms found by graph tools [15] remain possible; they lack the spatial weight-sharing interpretation that motivates equivariant architectures, and we leave them open.)

5 Honeycomb Floquet code: circuit-level exactness

5.1 Trivalence removes the schedule obstruction

Because the honeycomb lattice is trivalent, every check is weight-2: each extraction is two entangling gates, schedule freedom is a single bit per edge ("which endpoint first"), and hook errors have weight ≤ 1 on data — they merge into existing single-qubit mechanism classes instead of creating orientation-dependent weight-2 classes. A uniform sublattice rule ("A-endpoint first") is automatically covariant under every sublattice-preserving lattice map. The contrast with the surface code's 4! candidate orders per plaquette accounts for the obstruction–exactness dichotomy of this paper.

5.2 Verified symmetry of the circuit-level DEM

With the schedule's period-3 color structure, the point group splits into color-preserving elements (pure spatial symmetries of the scheduled circuit), color-cycling elements (exact only when composed with a ± 1 sub-round shift — spacetime *screws*), and color-transposing elements (paired with time reversal; classified, untested). Verified at $L = 6$ ($R = 9$, 276 detectors, $p = 0.008$) by full-DEM multiset equality: translations $T(1, 1)$, $T(2, 2)$ and C_3 about a face are exact over the full DEM (4,020 groups) in *all three* extraction modes including naive; the C_3-about-a-vertex screw is exact in the time bulk, with the init transient pinned to absolute sub-round 7 excluded by a bulk margin (the temporal analogue of a spatial boundary). The orbit-engineered schedule equals the naive one everywhere — we record this because the fact that engineering is unnecessary is itself part of the result.

5.3 Logical observables, the termination effect, and $GL(2, \mathbb{F}_2)$

To decode we need observables. We *derive* them rather than construct them: sampling the noiseless circuit and taking the $\mathbb{F}_2$ null space of the record matrix yields every deterministic record parity; quotienting by the detector span leaves the trackable logicals. This procedure produced two findings:

Termination. With $|+\rangle^{\otimes n}$ initialization, the deterministic wrap pair exists only when the

memory is terminated one sub-round after a complete period, with transversal readout in the Y basis ($S \equiv 1 \bmod 3$); X- and Z-basis terminations carry zero deterministic wraps. This is the honeycomb code's logical-type rotation [10] appearing as an operational constraint, found empirically by sweeping the termination phase. The surviving observables are "moving logical" spacetime tubes — at $L = 6$, weights 116 and 128 records, of which only 8 and 14 are data records — and the experiment must be terminated with the right structural detectors (initialization-round checks, virtual-round closures reconstructing the missing color from data pairs, and final window-vs-data comparisons) for the memory to have no undetectable boundary errors; the construction is gate-checked by noiseless determinism throughout (Appendix C).

The logical action. Solving (A, X) for the nine-element group $p3_9 = \{C_3^a\, T(2,2)^b\}$ (a closed abelian subgroup of $p3$ at $L = 6$; C_3 and $T(2,2)$ commute because $R(2,2) \equiv (2,2) \bmod 6$):

element class	A	twist
T_{22}, T_{44}	I	χ-weights up to 168
C_3, C_3T	$\left[\begin{smallmatrix}1&1\\1&0\end{smallmatrix}\right]$ (order 3)	nonzero
C_3^2, C_3^2T	$\left[\begin{smallmatrix}0&1\\1&1\end{smallmatrix}\right] = \left[\begin{smallmatrix}1&1\\1&0\end{smallmatrix}\right]^2$	nonzero

verified against all 4,584 mechanism groups of the decoding circuit ($p = 0.002$, naive extraction, observables included). None of this is specific to $L = 6$: the construction gates reproduce the same action at $L = 3$, and at $L = 9$ (810 detectors, 10,314 mechanism groups, $T(3,3)$ replacing $T(2,2)$ for closure) the termination effect reproduces (exactly two deterministic wraps, again only at $S \equiv 1 \bmod 3$ with Y readout), C_3 solves with the identical order-3 matrix, and the nine-element closure verifies in full. Translations preserve the homology classes; C_3 **mixes the two logical qubits**, consistent with the automorphism picture of Floquet codes [3, 14] — here appearing not as a logical gate to be exploited but as a constraint the decoder must respect. A scalar-label equivariant decoder cannot express this symmetry; the correct model is four-class and jointly equivariant,

$$f(x)[c] \;=\; \frac{1}{|G|}\sum_{g} \mathrm{net}(\tilde{x}_g)[\, A_g\, c \,\oplus\, X_g \cdot \tilde{x}_g\,], \qquad \tilde{x}_g = P_g^{-1}\text{-permuted } x, \tag{1}$$

which is exactly invariant on realizable syndromes (measured deviation 7.5×10^{-9}; the gauge caveat of §3.1 is why the probe uses sampled shots).

6 Learning experiments

Protocol common to all experiments: matched parameter counts across arms (the tied models share one backbone evaluated $|G|$ times — parameters are matched, FLOPs are not, noted where relevant), Adam with learning rate 10^{-3} and batch size 256, a fixed budget of 4,000 steps (20,000 for the large-n and scale arms), 5 random seeds (3 for the large-n extension, 2 for the scale ablation; uncertainties are standard deviations across seeds), 200,000-shot test sets, accuracy = exact match of the observable mask (joint over both logicals where two exist). Baselines: MWPM via pymatching on the decomposed DEM, and the majority class. Total compute is modest: all MLP experiments ran on one consumer GPU (GTX 1070), the CNN sweeps on single A10G instances; the full experimental record behind this paper is a few tens of GPU-hours. These are inductive-bias measurements at matched parameters, not decoder benchmarks: we report accuracy at fixed noise rates — the evidence class of The END's code-capacity study [5] — rather than per-round logical-error or threshold curves, which would require competitive backbones (§7); §6.2 includes a noise-rate robustness sweep.

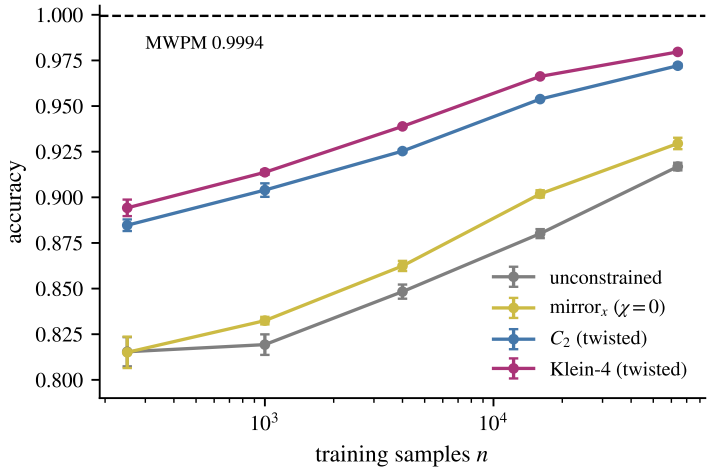


Figure 3: **The twisted elements carry the benefit** (surface code $d = 5$, phenomenological noise, matched parameters, 5 seeds). The $\chi = 0$ mirror tracks the unconstrained baseline; the twisted C_2 and Klein-4 arms dominate at every n — Klein-4 at 4,000 samples exceeds unconstrained at 64,000 ($> 16\times$). Data: `runs/equiv_decoder_d5.json`, `_d5b.json`.

6.1 Surface code, phenomenological noise: twisted elements carry the value

$d = 3$ (memory-Z, $p = 0.008$, MWPM 0.9948; 5 seeds): the C_2-twisted-invariant decoder beats plain at every n — $0.9791 \pm .004$ vs $0.9371 \pm .007$ at $n = 250$ (+4.2 pp), reaching $0.9943 \pm .0002$ at 64,000 against MWPM 0.9948, where plain lags (0.9908); the twisted model at 16,000 already exceeds plain at 64,000 (0.9921 vs 0.9908), $\approx 4\times$ sample efficiency.

$d = 5$ (MWPM 0.9994), element ablation at matched everything:

n	plain	mirror$_x$ (χ=0)	C_2 (twisted)	Klein-4
250	0.8154	0.8151	0.8847	**0.8942**
4,000	0.8483	0.8624	0.9253	**0.9389**
64,000	0.9169	0.9295	0.9721	**0.9797**

Klein-4 at 4,000 exceeds plain at 64,000: $> 16\times$. The mirror that *fixes* the logical representative ($\chi = 0$) contributes almost nothing; the elements that move it carry the benefit. Our reading: the twist supplies exactly the nonlocal label-consistency structure the decoder otherwise has to infer from data — the twist is not a nuisance but the signal. At 1M samples and a $5\times$ longer step budget Klein-4 holds its lead (0.9951 vs 0.9863) and closes to 0.4 pp of MWPM. Capacity limits, reported in full: the 2-layer MLP family saturates by $d = 7$ (plain falls below majority; Klein-4 rescues +15 pp but plateaus), fails entirely at $d \geq 9$, and a 3D-CNN backbone moves the $d = 7$ ceiling to $\approx$0.925 at 16k samples with the same plateau thereafter. Correlated pair-noise grows the symmetry advantage (+11 pp at 64k) but does not let any arm beat MWPM here; we claim no MWPM superiority anywhere in this paper.

6.2 Honeycomb code, circuit-level noise: a learnability switch

$L = 6$, $R = 9$, MY termination, naive extraction, $p = 0.002$ (noise gate: p chosen by halving from 0.008 until MWPM joint accuracy lands in a learnable band; final MWPM 0.9471, majority 0.2908, flip rate 0.71 — a well-mixed four-class task). Matched 63,236-parameter MLP backbone:

n	plain	C_3-tied (3 elements)	$p3_9$-tied (9 elements)
250	$0.2575 \pm .005$	$0.2820 \pm .004$	$\mathbf{0.3633 \pm .012}$
1,000	$0.2665 \pm .003$	$0.3021 \pm .004$	$\mathbf{0.4037 \pm .006}$
4,000	$0.2721 \pm .001$	$0.3226 \pm .004$	$\mathbf{0.4698 \pm .003}$
16,000	$0.2741 \pm .000$	$0.3488 \pm .001$	$\mathbf{0.6005 \pm .009}$
64,000	$0.2899 \pm .002$	$0.5145 \pm .004$	$\mathbf{0.7443 \pm .004}$

The unconstrained network never leaves the majority baseline. With 4× more data and 5× more steps it finally departs from it ($0.3907 \pm .001$ at $n = 256{,}000$) — at which point the tied model is at $0.8060 \pm .003$, and the tied model at $\mathbf{n = 1{,}000}$ (0.4037) **still exceeds plain at n = 256,000**: a $\geq 256\times$ sample-efficiency separation, against $\approx 4\times$ (d=3) and $> 16\times$ (d=5) where the exact symmetry lives below circuit level. On the code whose circuit-level DEM is exactly symmetric, equivariance is the difference between not learning and learning — the circuit-level sharpening of The END's untwisted-pooling ablation.

To exclude a capacity artifact, we ran a pre-registered scale ablation (predictions logged in the script before the run): at 7.1× the parameters ($h = 512$; 449,540) and 22.5× ($h = 1024$; 1,423,364) the unconstrained MLP still sits at the majority baseline at $n = 64{,}000$ ($0.2858 \pm .001$ and $0.2862 \pm .001$ vs majority 0.2908), and at $n = 256{,}000$ reaches only $0.3602 \pm .001$ — no better than the narrow model. Width does not help the tied model either: $p3_9$ at $h = 512$ gives 0.7386/0.8007 at 64k/256k versus 0.7443/0.8060 at $h = 128$. Together with the convolutional backbone swap of §6.4, this identifies the residual gap to MWPM as a property of the architecture class, not of parameter count or tying strategy.

Noise-rate robustness. A pre-registered sweep at $p \in \{0.001, 0.002, 0.004\}$ (fresh 200k test sets per rate, 3 seeds) bounds the switch claim in both directions:

p	MWPM	majority	plain@4k	$p3_9$@4k	plain@64k	$p3_9$@64k
0.001	0.9956	0.4086	0.3993	**0.8544**	0.6353	**0.9625**
0.002	0.9471	0.2908	0.2717	**0.4676**	0.2890	**0.7434**
0.004	0.6226	0.2536	0.2501	0.2544	0.2511	0.2615

The tied-over-plain ordering holds in every (p, n) cell, MWPM is unbeaten everywhere, and the solved (A, X) action is bit-identical across rates, as its structural origin requires. The registered prediction (strict switch at every rate) is partially violated at both ends, and we report it as such: at $p = 0.001$ the plain model escapes the baseline at 64,000 samples ($0.6353 \pm .001$) while remaining 33 pp behind the tied model, whose 4,000-sample accuracy (0.8544) still exceeds plain at 64,000 ($> 16\times$); at $p = 0.004$ neither arm escapes baseline at these sample sizes (tied $0.2615 \pm .001$ vs majority 0.2536, and MWPM itself falls to 0.62). The strict switch — plain pinned at the majority baseline through 64,000 samples — is a property of the gated rate; the sharper statement the sweep supports is that exact tying **widens the noise window in which the task is learnable at all** at fixed capacity and data.

Limitations: $p3_9$ at 256k remains 14.1 pp below MWPM (architecture class, not data or width — the curve is unsaturated and 4× width does not move it); 20k-step training at fixed $n = 64$k mildly overfits relative to 4k steps on the tied arms (0.7306 vs 0.7443); the scale arms use 2 seeds and the noise sweep 3, at $n \in \{4\text{k}, 64\text{k}\}$ only.

6.3 What the comparison across codes says

Identical backbones, identical training protocol, comparable detector counts (240–588 vs 360): the surface-code architecture benefits from symmetry below circuit level and *cannot* be given

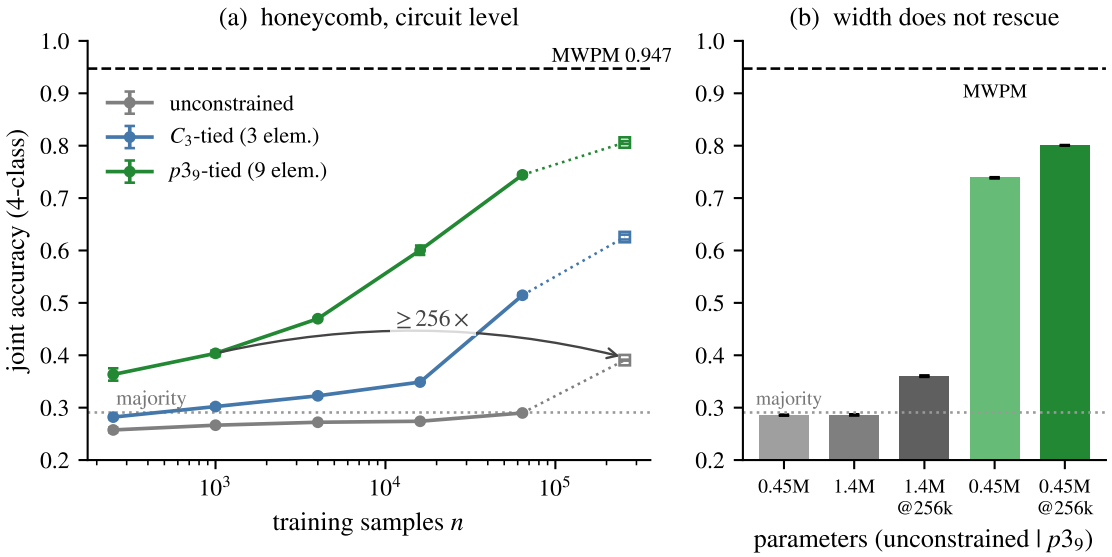


Figure 4: **Equivariance as a learnability switch** on the honeycomb circuit-level task ($L = 6$, $p = 0.002$, MWPM 0.9471, majority 0.2908). (a) Matched 63k-parameter backbones, 5 seeds; open squares are the 256k extension (3 seeds, 20k steps). The unconstrained MLP stays at the majority baseline through 64,000 samples; the $p3_9$-tied model reaches 0.806 at 256,000, and at $n = 1{,}000$ already exceeds the unconstrained model at $n = 256{,}000$ (arrow): $\geq 256\times$. (b) Pre-registered scale ablation: 7×–22.5× more parameters do not wake the unconstrained model, and 4× width does not move the tied model — the residual MWPM gap is architecture class, not capacity. Data: `runs/honeycomb_decoder{,_bign,_scale}.json`.

exact symmetry at circuit level (§4); the honeycomb architecture is handed exact circuit-level symmetry by the geometry and converts it into the largest separations we measure. The lattice that buys nothing as a learned-model *input geometry* (our long-running negative results on hexagonal inductive bias elsewhere) buys everything as the generator of the *code's* symmetry group.

6.4 Composition with convolutional translation tying — and what weight sharing cannot do

We replace the MLP backbone with a 46,644-parameter 3D CNN over the (time, face$_i$, face$_j$) detector grid — circular padding on the torus dimensions, so the convolution is natively translation-equivariant — and compose it with the same exact 4-class twisted averaging (invariance probe 7.5×10^{-9}). Three questions (5 seeds, same protocol):

n	CNN (plain)	CNN + C_3 (3 el.)	CNN + $p3_9$ (9 el.)	MLP + $p3_9$ (ref.)
250	0.2555 ± .005	0.2810 ± .009	**0.3647 ± .011**	0.3633
1,000	0.2649 ± .003	0.3021 ± .008	**0.4337 ± .012**	0.4037
4,000	0.2732 ± .002	0.3393 ± .006	**0.5291 ± .008**	0.4698
16,000	0.2820 ± .003	0.4055 ± .007	**0.6351 ± .013**	0.6005
64,000	0.3184 ± .007	0.5380 ± .008	**0.7502 ± .015**	0.7443

First, the translation prior alone does *not* rescue learnability: the unconstrained CNN sits at or near the majority baseline through 16,000 samples and reaches only 0.318 at 64,000. Second, the composition pays where data is scarce: CNN+$p3_9$ exceeds MLP+$p3_9$ by +3.0/ +

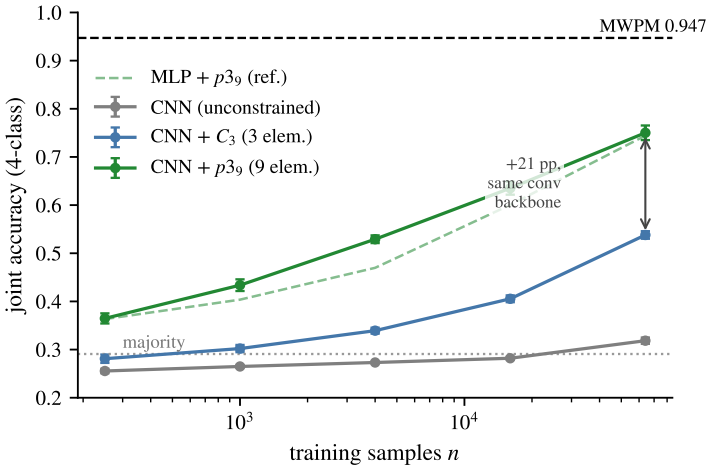


Figure 5: **Convolutional weight sharing cannot absorb the label twist** (honeycomb task, 46,644-parameter 3D-CNN backbones, 5 seeds). The translation-equivariant CNN alone stays near majority; composing it with the exact twisted averaging pays +3–6 pp over the tied MLP where data is scarce; and on the same conv backbone the full nine-element group adds +21 pp at 64,000 over C_3-only averaging — the twist is a cocycle on the output, invisible to feature weight sharing. Data: `runs/honeycomb_decoder_cnn.json`.

5.9/ + 3.5 pp at n = 1,000/4,000/16,000, converging by 64,000 (0.750 vs 0.744); the ≈0.75 ceiling at 64,000 — some 20 pp below MWPM — is unmoved by the backbone change, identifying the remaining gap as architecture class rather than tying strategy (and not raw width — §6.2's scale ablation), consistent with the surface-code $d = 7$ CNN plateau (§6.1). Third, on the *same convolutional backbone*, going from C_3-only averaging to the full nine-element group adds +21 pp at 64,000 — even though convolutions are already translation-equivariant. The reason is that the label twist under translations ($A = I$ but $\chi \neq 0$) is a cocycle on the *output*, invisible to feature-level weight sharing: convolution can share features across translated syndromes, but only the twisted average makes the *label* transform correctly. Weight sharing handles the features; the twist must be handled explicitly. Together with The END's pooling ablation and the element ablation of §6.1, this isolates a design rule for equivariant decoders: the syndrome-dependent label correction is not an implementation detail of the symmetry — it is the component that carries the benefit.

7 Discussion

What we claim. (i) Exact twisted equivariance of the decoding problem is a circuit-level property that depends on the code, decidable by the DEM solver of §3; (ii) for the surface code it is obstructed structurally, not for want of schedule engineering; (iii) for the honeycomb Floquet code it holds exactly with zero engineering, the logical action includes a $GL(2, \mathbb{F}_2)$ mixing under C_3, and exact tying produces order-of-magnitude-plus sample-efficiency separations, up to a learnability switch.

What we do not claim. No decoder here beats MWPM; the backbones are small; one code size per learning claim (the honeycomb structural results are verified at $L = 3$, 6, and 9); uniform noise rates; torus boundary conditions for the honeycomb results (planar boundaries break translations for any code); the C_2/time-reversal class of the honeycomb code is classified but untested. Novelty is claimed "to our knowledge" against the positioning in §1 and the related work below; the formulation layer is The END's.

Why it might matter. Real-device syndrome data is scarce — the regime where 4× to ≥ 256× sample efficiency is the difference between a trainable decoder and an untrainable one. If learned decoders are part of the fault-tolerance stack, then "does the code hand its symmetry to the learner at circuit level" is a code-selection criterion that has not, to our knowledge, appeared in the Floquet-versus-surface conversation. The honeycomb code's known costs (dynamic logicals, measurement overhead) and benefits (weight-2 checks, hardware fit) acquire one more entry on the benefit side. Directions we consider promising: schedule-aware non-affine group actions; the duality element on torus codes; DEM-automorphism mining [15] given a twist-aware extension; and real-device data, where the AlphaQubit results [1] indicate learned decoders are already the accuracy frontier.

Related work

The END [5] — twisted equivariance, toric code, code capacity (the formulation this paper extends to circuit level). AlphaQubit [1] — learned decoding at circuit level and on device data, no exact equivariance. Automorphism ensemble decoding [15] — DEM automorphisms via graph tools for ensemble BP on bivariate-bicycle codes; permutes syndromes only (no logical action/twist), no geometric analysis, no learning. Hook-error schedule design [12, 6] — schedule choice for circuit distance; a different objective from DEM symmetry (§4). Honeycomb code [10, 8, 9] — construction, benchmarks, boundaries. Floquet/dynamic automorphism codes [3, 14] — automorphisms as logical gates; our C_3 result is the decoder-side face of the same structure. Group-equivariant networks [2] — the general averaging principle; The END's σ-dependent output representation, and our (A, X) extension of it, sit outside the standard fixed-representation theory.

Code availability

The DEM symmetry solver and the twisted-equivariant wrapper are released as the `demsym` package (`pip install demsym`), which ships the toric-code validation of §3.2 as a test; experiment scripts and the logged runs behind every number in this paper accompany the repository.

A Solver details

For a DEM with detector-set groups $\{D \mapsto \text{multiset of } (o, p)\}$ and a candidate (P, A): applying A to the source multiset and comparing against the target group $P(D)$ constrains the offset bit-vector $b(D) \in \mathbb{F}_2^k$ to the consistent set $B(D) = \{b : \{(Ao \oplus b, p)\} = \text{by_v}[P(D)]\}$. If $B(D) = \emptyset$ for any D, the candidate is rejected. For each observable bit i with a forced value ($|\{b_i : b \in B(D)\}| = 1$), the row $\mathbf{1}_{P(D)}$ joins a linear system $X_i \cdot \mathbf{1}_{P(D)} = b_i$ over $\mathbb{F}_2$, solved by Gaussian elimination (free variables set to zero — the gauge choice). A final pass re-verifies the full multiset equality for every group under the solved (P, A, X), which makes the result independent of the gauge and of which groups happened to force equations. Candidate A's are enumerated over $GL(k, \mathbb{F}_2)$ for $k \leq 3$ (orders 1, 6, 168); for larger k a candidate list is supplied (the toric validation passes the 24 observable permutation matrices). Complexity is $O(|\mathcal{A}| \cdot \sum_D |B\text{-search}| \cdot 2^k)$ plus one elimination per bit; every instance in this paper solves in seconds. The per-group-multiset formulation (rather than per-mechanism pairing) is load-bearing: stim merges mechanisms with identical (detectors, observables) support, and probability collisions across a candidate map generate false per-mechanism equations from $d = 5$ onward.

B Surface-code null searches

Affine family. Candidates are $P(x, y, t) = (\sigma(x, y), \pm t + \Delta t)$ with $\sigma \in \{\text{mirror}_x, \text{mirror}_y, C_2, \text{id}\}$ composed with spatial shifts, time reversal, and temporal shifts $\Delta \in [-2, 2]^3$, mapped onto detector coordinates of stim's unrotated memory circuits; maps that fail to permute the detector-coordinate set are discarded before solving, leaving 999 candidates per distance. Verified count: 0 at $d = 3$, 0 at $d = 5$ (`runs/circuit_symmetry_mine.json`).

Schedule co-design. All counts from `runs/schedule_codesign.json`:

family	tried	collision-free	pass G1–G3	pass G4
A (uniform per-orientation)	576	96	16	0
B (mirror-paired halves)	576	0	0	0
C (mirror-paired, time-reversed half)	576	32	8	0

The mirror-axis argument. Label the axis ancilla's four data neighbors N, E, S, W; the mirror τ fixes N and S and exchanges E $\leftrightarrow$ W, and maps the axis ancilla to itself.

Lemma. *No extraction order $\pi = (\pi_1, \pi_2, \pi_3, \pi_4)$ of $\{N, E, S, W\}$ has a τ-invariant hook-error support family.*

Proof. A hook error after step k propagates to the suffix set $S_k(\pi) = \{\pi_{k+1}, \ldots, \pi_4\}$, so the axis ancilla contributes mechanisms with data supports S_1, S_2, S_3 (and the trivial S_0). Under the mirror these mechanisms must map to mechanisms of the same ancilla, so the family $\{S_1, S_2, S_3\}$ must be τ-invariant; since the $|S_k| = 4 - k$ are distinct, each S_k must be τ-invariant individually. $S_3 = \{\pi_4\}$ forces $\pi_4 \in \{N, S\}$. Then $S_2 = \{\pi_3, \pi_4\}$: if $\pi_3 \in \{E, W\}$, $\tau(S_2)$ contains the opposite of π_3 and $\tau(S_2) \neq S_2$; so $\pi_3 \in \{N, S\}$ and $S_2 = \{N, S\}$. Then $\pi_2 \in \{E, W\}$ and $S_1 = \{\pi_2, N, S\}$, whose image $\tau(S_1)$ contains the opposite of $\pi_2 \notin S_1$ — contradiction. □

The same argument applied to prefix sets rules out the time-reversed pairing of family C at the axis.

C Honeycomb memory construction

Detectors are built from four structural families: (i) initialization-round checks — each $s = 0$ XX edge record is individually deterministic on $|+\rangle^{\otimes n}$, and the $k = 0$ inference window of every color-0 face is deterministic (the face stabilizer is X-type); (ii) bulk window comparisons — consecutive inference windows of each face; (iii) virtual-round closures — at the final transversal readout (basis of the next, unexecuted sub-round), color-0 edge values are reconstructed from data-record pairs to close the in-flight windows of faces missing that color; (iv) final window-vs-data comparisons for faces whose color matches the readout basis. Gate G2 (noiseless determinism of every detector and observable) holds throughout; observables are derived as the $\mathbb{F}_2$ null space of the noiseless record matrix modulo the detector span. Termination sweep at $L = 6$: MX readout, 0 deterministic wraps; MY, 2 wraps (weights 116 and 128) plus one zero-data-support parity; MZ, 0. The zero-data parity (the global product of the last XX round) is dropped rather than declared: a global-support detector turns every boundary-layer error into an undecomposable hyperedge for matching while carrying essentially no decoding signal. Sampling note: the empirical null space must be sampled past the record matrix's random rank (4,096 noiseless shots at $L = 9$, where the default 1,024 under-samples and reports spurious wraps — caught by the pre-registered gate).

D Provenance: claim → logged run

Every number in this paper traces to a completed, logged run; the audit script `scripts/preprint_audit.py` re-checks 94 of the numeric claims against the JSON outputs (94 pass, 0 fail at submission).

claim	file
toric bridge	`runs/toric_end_bridge.{log,json}`
999-map null	`runs/circuit_symmetry_mine.{log,json}`
1,728-schedule null	`runs/schedule_codesign.{log,json}`
honeycomb DEM symmetry	`runs/honeycomb_code_symmetry.{log,json}`
(A, X) table, gates $(L = 3, 6)$	`runs/honeycomb_decoder_gates.{log,json}`
surface learning $d = 3/5/7$, big-n, correlated, CNN	`runs/equiv_decoder*.{log,json}`
honeycomb learning	`runs/honeycomb_decoder.{log,json}`
honeycomb big-n	`runs/honeycomb_decoder_bign.{log,json}`
honeycomb CNN	`runs/honeycomb_decoder_cnn.json`
scale ablation	`runs/honeycomb_decoder_scale.{log,json}`
structure at $L = 9$	`runs/demsym_honeycomb_L9.{log,json}`
noise-rate sweep	`runs/honeycomb_decoder_psweep.{log,json}`

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